Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building cross-platform software, automated tests, APIs, and performance-sensitive systems. Strong CS fundamentals across C/C++, Python, Java, Linux, operating systems, data structures, concurrency, Git, CI/CD, debugging, and simulation-minded development for embedded and mission-critical software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built SQL-backed platform workflows for **500+** users, translating requirements into APIs, validation logic, background jobs, and release tasks for reliable cloud-hosted systems.
- Improved runtime performance by optimizing indexed access paths and asynchronous jobs, reducing dashboard latency from **1.2s to 350ms** while strengthening troubleshooting handoffs.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Python and JavaScript services handling **10,000+** daily requests, implementing REST APIs, concurrent file processing, and model-backed execution workflows.
- Debugged OCR, database, file-ingestion, and LLM execution failures, documenting root causes, test cases, and recovery steps that reduced feedback turnaround by **60%**.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Created Python evaluation pipelines over **100,000+** records, producing reproducible metrics, logs, and debugging artifacts for model behavior analysis and data validation.
- Collaborated with researchers and engineers to convert experiments into maintainable scripts, reports, and issue summaries, supporting iterative AI/ML software development.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservices and ETL workflows using Python, OOP patterns, AWS Glue, and S3, improving API response time by **30%**.
- Maintained Jenkins pipelines and **20+** PyTest/Mockito suites across Windows, Linux, and macOS validation flows, increasing CI reliability by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built modular services with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting **2M+** synchronized records across distributed product workflows.
- Automated Docker and Jenkins build/test flows, cutting release time by **50%** and improving repeatable deployments, debugging, and documentation for engineering teams.

Skills

Systems: Operating systems, Linux/Unix, command line, concurrency, memory management, process scheduling, networking basics, embedded Linux familiarity
Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL, Bash; Rust/Zig interest and ramp-up readiness
Embedded & Protocols: RTOS concepts, device-driver concepts, TCP/IP, UDP, SPI/I2C/CAN/RS422/RS485 familiarity, simulators, test harnesses
Delivery: Git, CI/CD, Jenkins, GitHub Actions, Docker, Linux debugging, logs, automated testing, documentation, Agile collaboration

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling algorithms with metric calculation and Gantt visualization across **1K+** process simulations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+** databases, working across Java services, TypeScript UI, Docker setup, schema exploration, and query execution.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a containerized REST data-processing API with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy.